

Adapted antisymmetric stochastic integrals: the Hilbert-transform constant

arXiv automatic math run `fa_banach_001`

2026-06-18

Status

This is a *literature-implied answer (partial constant)* for the antisymmetric route in Remark 5.12 of Yaroslavtsev's *Fourier multipliers and weak differential subordination of martingales in UMD Banach spaces*, arXiv:1703.07817v5 [1].

The source remark asks for the analogue of Theorem 5.10 when the operator A on the Hilbert noise space is antisymmetric. Such an estimate with a constant linear in $\beta_{p,X}$ would imply the linear Hilbert-transform estimate

$$\|\mathcal{H}_X\|_{L^p(\mathbb{R};X) \rightarrow L^p(\mathbb{R};X)} \leq C_p \beta_{p,X}.$$

Identification

Let W^H be an H -cylindrical Brownian motion, let Φ be an adapted stochastic-integrable process with values in $\mathcal{L}(H, X)$, and let

$$M = (\Phi \cdot W^H), \quad N = ((\Phi A) \cdot W^H),$$

where $A^* = -A$ and $\|A\| \leq 1$. For every $x^* \in X^*$,

$$[\langle M, x^* \rangle, \langle N, x^* \rangle]_t = \int_0^t \langle \Phi(s)^* x^*, A\Phi(s)^* x^* \rangle_H ds = 0,$$

so M and N are orthogonal in the sense of Osekowski–Yaroslavtsev [2]. Also

$$[\langle N, x^* \rangle]_t = \int_0^t \|A\Phi(s)^* x^*\|_H^2 ds \leq \int_0^t \|\Phi(s)^* x^*\|_H^2 ds = [\langle M, x^* \rangle]_t,$$

so N is weakly differentially subordinate to M .

Theorem 2.1 of [2], applied with $\Phi_0(x) = \Psi_0(x) = \|x\|^p$, gives

$$\mathbb{E}\|N_t\|^p \leq \hbar_{p,X}^p \mathbb{E}\|M_t\|^p, \quad t \geq 0,$$

where $\hbar_{p,X}$ is the norm of the Hilbert transform on $L^p(\mathbb{R}; X)$. The same consequence is restated in the introduction of [3]: orthogonal weakly differentially subordinate martingales satisfy the sharp $\hbar_{p,X}$ -bound.

Scope and obstruction

This does *not* fully solve Remark 5.12 in the beta-constant form suggested by Theorem 5.10. It proves the adapted antisymmetric estimate with $\hbar_{p,X}$, not with $\beta_{p,X}$. Osekowski–Yaroslavl'tsev explicitly record that the case $\Phi_0 = \Psi_0 = \|\cdot\|^p$ is tied to the celebrated open problem whether $\hbar_{p,X} \lesssim_p \beta_{p,X}$; the best general upper bound available there is still quadratic, $\hbar_{p,X} \leq \beta_{p,X}^2$.

Therefore the current status is:

$$\|((\Phi A) \cdot W^H)_t\|_{L^p(\Omega;X)} \leq \hbar_{p,X} \|(\Phi \cdot W^H)_t\|_{L^p(\Omega;X)}$$

for adapted antisymmetric A , while the desired promotion to

$$\|((\Phi A) \cdot W^H)_t\|_{L^p} \leq C_p \beta_{p,X} \|(\Phi \cdot W^H)_t\|_{L^p}$$

would require resolving or bypassing the remaining Hilbert-transform/UMD constant comparison problem in this route.

References

- [1] I. Yaroslavl'tsev. *Fourier multipliers and weak differential subordination of martingales in UMD Banach spaces*. arXiv:1703.07817v5, 2017.
- [2] A. Osekowski and I. Yaroslavl'tsev. *The Hilbert transform and orthogonal martingales in Banach spaces*. arXiv:1805.03948, 2018.
- [3] I. Yaroslavl'tsev. *On strongly orthogonal martingales in UMD Banach spaces*. arXiv:1812.08049, 2018.